

SatQuery AI

Bi-Temporal Satellite Image Change Detection & Geospatial VQA Platform

COMPLETE SYSTEM ARCHITECTURE & JUDGES PRESENTATION DOSSIER

Target Audience: Project Team, Pitch Judges, Technical Evaluators

Document Version: 2.0 (Full 6-Phase Pipeline Coverage)

Core Capabilities: Multi-spectral GeoTIFF Ingestion, Optical-SAR Cross-Modal Fusion, Rule-Based & ML Change Understanding, Audit-Grade Evidence Extraction, Mission Control UX

Date: September 2026 | Status: Production Ready (130/130 Tests Passing)

EXECUTIVE OVERVIEW FOR JUDGES & EVALUATORS

SatQuery AI solves the critical challenge of analyzing bi-temporal satellite image pairs (T1 pre-change vs T2 post-change). Unlike traditional black-box vision systems, SatQuery AI combines **multi-spectral spectral indices (NDVI, NDWI, NDBI)**, **SAR radar backscatter fusion**, and **explainable audit trails** to identify where changes occur, what land-cover transition happened (e.g., Deforestation, Urban Expansion, Demolition), and why the AI reached that conclusion with 100% mathematical trace transparency.

1. Executive Summary & Value Proposition

Modern satellite intelligence relies heavily on Earth Observation (EO) data from optical (Sentinel-2, Landsat) and Synthetic Aperture Radar (SAR / Sentinel-1) constellations. However, defense, environmental monitoring, and urban planning teams face three massive bottlenecks when analyzing bi-temporal (T1 vs T2) imagery:

- **The Black-Box Reliability Gap:** Deep learning models predict change masks but fail to provide auditability or explain why a cluster of pixels was flagged as urban expansion vs seasonal vegetation drop.
- **Multi-Modal Blind Spots:** Optical sensors cannot penetrate cloud cover or rain, while SAR sensors operate through all weather but suffer from complex speckle noise. Fusing them seamlessly requires deterministic cross-modal validation.
- **Heavy Geospatial Format Complexity:** Raw GeoTIFF rasters contain non-standard Coordinate Reference Systems (CRS), multi-band combinations (RGB, NIR, SWIR), and massive pixel dimensions that crash conventional web apps.

The SatQuery AI Solution:

SatQuery AI provides an end-to-end full-stack platform featuring windowed raster ingestion, a modular multi-model registry, multi-spectral spectral index delta engines, SAR backscatter cross-validation, and an interactive Mission Control dashboard. Every analysis returns an explicit **Execution Trace**, zero-hallucination structured evidence metrics, and interactive spatial bounding boxes.

Key Differentiators Summary Table

Feature / Metric	Traditional GIS / Standard AI	SatQuery AI Platform
Explainability	Black-box probabilities; no audit trail	Full Execution Trace + Feature Deltas (NDVI, NDWI, NDBI)
Sensor Modalities	Optical-only OR SAR-only in isolation	Fused Optical-SAR Cross-Modal Engine with Agreement Scores
Raster Processing	Full image loading into memory (OOM crash)	Windowed GDAL/Rasterio reading & downsampled previews
Evidence Extraction	LLM text generation prone to hallucination	Deterministic Math & Rule Engine (Zero Hallucination)
Testing & Quality	Ad-hoc script execution	130/130 Automated Pytest Suite covering edge cases

2. Full System Architecture & 6-Phase Pipeline

SatQuery AI follows a decoupled micro-architecture separating raster data processing, model inference adapters, agentic orchestration, and the Next.js frontend user interface.

End-to-End Pipeline Workflow:

Phase 1 & 2: Ingestion Engine	Rasterio parsing, CRS metadata validation, windowed reading, PNG thumbnail generator.
Phase 3: Change Detector	Bi-temporal pixel matrix subtraction, contour grouping, spatial bounding box extraction.
Phase 4: Change Understander	SpectralLandCoverClassifier (T1/T2), NDVI/NDWI/NDBI delta calculation, transition taxonomy.
Phase 5: Evidence Engine	Spatial, Spectral, Statistical, Temporal evidence categorization & contradiction detection.
Phase 6: Orchestrator & SAR	NLP intent routing, Optical-SAR cross-modal fusion (Agreement / Conflict analysis).
Phase 7: Adaptation & Benchmarks	BigEarthNet ResNet-50 adaptation, RSVQA & VRSBench benchmark evaluation pipeline.

3. Technical Phase Deep Dives

Phase 1 & 2: Ingestion, Metadata & Model Registry

The Ingestion Service processes GeoTIFF rasters with full support for multispectral bands (Red, Green, Blue, Near-Infrared / NIR, Short-Wave Infrared / SWIR). Using rasterio, it extracts spatial metadata: Coordinate Reference System (e.g. EPSG:4326), bounding box coordinates, pixel resolution (m/px), and band data types. Memory overhead is minimized through windowed sampling during thumbnail PNG generation.

- **Lazy-Loading Model Registry:** Models (adapted VLM, change detectors, land-cover classifiers) are registered with lazy initialization, ensuring zero VRAM usage until an inference query arrives.
- **CPU Fallbacks:** Gracefully handles environments without GPU CUDA support, executing NumPy optimized matrix operations.

Phase 3 & 4: Change Detection & Spectral Taxonomy Understanding

Phase 3 isolates change region masks between pre-change image T1 and post-change image T2. Phase 4 then inspects each bounding box and runs the SpectralLandCoverClassifier on T1 and T2 pixel matrices.

Spectral Indices Formulae:

- **NDVI (Normalized Difference Vegetation Index):** $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$ — Quantifies vegetation health & biomass.
- **NDWI (Normalized Difference Water Index):** $(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$ — Highlights open water bodies.
- **NDBI (Normalized Difference Built-Up Index):** $(\text{SWIR} - \text{NIR}) / (\text{SWIR} + \text{NIR})$ — Identifies artificial structures & concrete.

Transition Taxonomy Rules:

Change Category	T1 Land Cover	T2 Land Cover	Spectral Trigger / Delta Signature
BUILT_UP_EXPANSION	Agriculture / Bare Soil	Built-Up	NDBI Increase ($> +0.05$), NDVI Decrease
DEMOLITION	Built-Up	Bare Soil	NDBI Decrease (< -0.05)
DEFORESTATION	Forest / Vegetation	Bare Soil	NDVI Severe Drop ($\Delta < -0.30$)
VEGETATION_GAIN	Bare Soil	Vegetation	NDVI Strong Surge ($\Delta > +0.25$)
WATER_BODY_CHANGE	Land / Water	Water / Land	NDWI Delta Shift ($ \Delta > +0.20$)
ROAD_DEVELOPMENT	Agriculture	Linear Built-Up	NDBI Rise + High Spatial Aspect Ratio

Phase 5: Audit-Grade Evidence Extraction & Contradiction Engine

Phase 5 is the verification layer. Rather than taking Phase 4 predictions as absolute truth, Phase 5 evaluates raw pixel statistics, spatial dimensions, and spectral deltas to calculate an independent evidence_score.

- **Contradiction Penalty:** If a region is classified as *Built-Up Expansion* but NDBI actually decreased, the engine penalizes the score (-0.2 per contradiction) and flags a pipeline warning.
- **Structured Categories:** Isolates evidence into Spatial (m² area, bbox), Spectral (NDVI/NDWI/NDBI deltas), Statistical (Mean/StdDev band ratios), Temporal (acquisition time delta), and Visual (crop coordinates).

Phase 6: Multi-Modal Optical-SAR Fusion & Orchestrator

SAR imagery (Sentinel-1) measures radar roughness and dielectric properties, providing cloud-penetrating surveillance. The RuleBasedOpticalSARAnalyzer validates spatial co-registration (CRS & transform matrices) and computes cross-modal agreement scores:

- **STRONG_AGREEMENT:** Optical NDBI high + SAR backscatter mean high → 100% confidence Urban Structure.
- **CONFLICT:** Optical NDWI indicates water but SAR backscatter is high → Flagged for analyst manual review (possible double-bounce scatter from flooded structures or oil slick).
- **OPTICAL_DOMINANT / SAR_DOMINANT:** Selected when cloud cover degrades optical bands, relying on SAR polarizations (VV, VH).

4. Frontend Mission Control & Full Technology Stack

The web interface is built using Next.js (React) and Vanilla CSS, featuring a sleek, dark-mode glassmorphic aesthetic tailored for military intelligence and geospatial analysts.

Frontend User Interface Features:

- **Dual-Canvas Bi-Temporal Viewer:** Side-by-side synchronized zoom & pan for T1 and T2 raster thumbnails with overlaid change bounding boxes.
- **Interactive Region Inspection Drawer:** Click any detected region box to view spectral index delta charts, land-cover transition pills, and raw band statistics.
- **Live Execution Trace Timeline:** Step-by-step telemetry stream displaying exact timestamped operations (e.g. `INGESTION_COMPLETED` → `CHANGE_DETECTION_COMPLETED` → `EVIDENCE_ATTACHED`).
- **Audit Evidence Breakdown Card:** Highlights supporting vs contradicting evidence with clear color-coded indicators.

Complete Technology Stack Overview

Layer	Technologies / Libraries Used	Role & Architecture Responsibility
Frontend UI	Next.js, React, Vanilla CSS, Lucide Icons	Responsive Mission Control dashboard, dual raster viewer, state management
Backend Framework	FastAPI, Uvicorn, Pydantic, Python 3.10+	RESTful API gateway, schema validation, async request routing
Geospatial & Rasters	Rasterio, GDAL, NumPy, PIL (Pillow)	GeoTIFF ingestion, CRS validation, windowed sampling, spectral index math
AI / ML Models	PyTorch, ResNet-50, Scikit-Learn	BigEarthNet multispectral adaptation, benchmark feature extractors
Testing & Quality	Pytest, Pytest-Asyncio, Coverage	130 automated unit & integration tests across all 6 phases

5. Step-by-Step Team & Judge Pitch Script (5-Min Masterclass)

Use this exact minute-by-minute script when presenting SatQuery AI to judges and evaluators during live demonstrations:

Time	Speaker Focus	Exact Script / Key Phrases to Say to Judges
0:00 - 1:00	The Hook & Problem Statement	"Judges, every day millions of satellite images are captured, but defense and environmental analysts face a critical bottleneck: black-box AI models flag changes without explaining why, and cloud cover blinds optical sensors. Today, we introduce SatQuery AI — an audit-grade bi-temporal geospatial platform that fuses multi-spectral optical and SAR radar imagery with 100% explainable execution traces."
1:00 - 2:30	Live Demo & Ingestion	"Watch how seamless this is. We upload a T1 pre-change GeoTIFF and a T2 post-change GeoTIFF. Our GDAL pipeline validates the CRS and streams downsampled thumbnails instantly. The Change Detection engine extracts change bounding boxes, while Phase 4 computes NDVI, NDWI, and NDBI spectral deltas to classify transitions like Deforestation or Urban Expansion."

2:30 - 3:30	Optical-SAR & Evidence Engine	<i>"What truly sets SatQuery AI apart is Phase 5 & 6. Notice this region: Optical NDBI indicates new construction, and our SAR engine validates it against radar backscatter, returning a STRONG AGREEMENT score. If a contradiction occurs — say, spectral indices disagree — our Evidence Engine penalizes the score and alerts the analyst."</i>
3:30 - 4:30	Execution Trace & Architecture	<i>"Every single step is tracked in our live Execution Trace log. No hallucinated LLM responses — strictly reproducible mathematical & spectral evidence. Our backend is backed by 130 passing Pytest unit tests, benchmarked against RSVQA and BigEarthNet datasets."</i>
4:30 - 5:00	Conclusion & Call to Action	<i>"SatQuery AI bridges the gap between raw Earth Observation data and auditable, high-confidence geospatial intelligence. Thank you, and we are ready for your questions!"</i>

6. Judge Q&A; Defense Cheat Sheet (Anticipated Technical Questions)

Be prepared to answer these exact technical questions from judges during the Q&A; session:

Q1: Is your system using deep learning or rule-based models?

Answer: SatQuery AI uses a **hybrid modular architecture**. Ingestion, spectral index deltas (NDVI/NDWI/NDBI), and contradiction evidence extraction use transparent, deterministic rule engines to guarantee auditability and zero hallucination. For high-level VQA and scene captioning, we integrate PyTorch ResNet-50 adapters fine-tuned on BigEarthNet multispectral datasets. All models inherit from pluggable Abstract Base Classes (ABC), allowing seamless ML drop-in replacements.

Q2: How do you handle cloud cover in optical satellite imagery?

Answer: That is why we built our **Phase 6 Optical-SAR Cross-Modal Fusion Engine**. When optical sensors are obscured by clouds (NDVI/NDWI data unavailable), the orchestrator identifies SAR_DOMINANT mode and uses Sentinel-1 Synthetic Aperture Radar backscatter polarizations (VV, VH), which penetrate clouds, rain, and night.

Q3: How do you prevent out-of-memory (OOM) crashes when uploading gigabyte-sized GeoTIFFs?

Answer: Our Phase 1 Ingestion engine utilizes rasterio windowed read blocks (decimated reading). Instead of loading the full 16-bit multi-gigabyte matrix into RAM, we sample bounding windows and downsample thumbnails for UI rendering while preserving full spatial resolution metadata (CRS, transform matrix, pixel bounds) for backend processing.

Q4: How do you prove that your evidence engine isn't hallucinating?

Answer: Phase 5 reuses the exact mathematical spectral index values (e.g. $\Delta\text{NDVI} = -0.42$) calculated during Phase 4. It does not use an ungrounded LLM generator. Furthermore, if spectral indices contradict the classified transition (e.g., NDBI drops during Urban Expansion), an explicit penalty (-0.2) is applied and a warning is logged in the **Execution Trace**.

Q5: What benchmarks have you evaluated your platform against?

Answer: We evaluate against **BigEarthNet** for multispectral land cover classification, as well as **RSVQA** (Remote Sensing Visual Question Answering) and **VRSBench** adapters located in our benchmarks/ codebase.